
HydDown CO2 Release — Technical Reference

Release 1.0

Anders Andreassen

Sep 11, 2026

CONTENTS

1	Introduction	3
1.1	Motivation	3
1.2	Scope	3
1.3	What the add-on provides	4
1.4	Architecture	4
2	Experimental Basis (CARDICE)	7
2.1	The vessel	7
2.2	Insulation and external heat transfer	8
2.3	Instrumentation	8
2.4	Test matrix	10
2.5	Raw-data alignment	12
3	Thermodynamics and Hand-overs	13
3.1	Why a second backend	13
3.2	Phase regimes	13
3.3	The self-consistent triple point	14
3.4	Hand-overs	14
3.5	Precomputed property tables	14
4	Discharge Model	15
4.1	Stagnation state	15
4.2	HEM mass flux	15
4.3	Non-equilibrium liquid discharge (HNE)	16
4.4	Orifice and discharge coefficient	16
5	Dry-Ice Estimation	19
5.1	Atmospheric (downstream) dry ice	19
5.2	In-vessel dry ice (opt-in)	19
5.3	Behaviour	21
6	Heat and Mass Transfer	23
6.1	Two-node wall above the triple point (NEM)	23
6.2	Natural convection	23
6.3	Below the triple point (two-zone wall)	23
6.4	Natural-convection gas-wall HTC (“calc”)	24
6.5	Gas-liquid interphase	24
6.6	Initialisation of the warm gas and split wall	24
6.7	Zone mass and energy balances (NEM)	25

6.8	The gas-temperature gap (stratification)	25
7	Validation	27
7.1	Methodology	27
7.2	Master comparison	27
7.3	Per-test comparisons	28
7.4	Benchmark against Vessfire (Test 5)	28
7.5	Sensitivity: wall-to-solid coefficient	28
7.6	Remaining gaps	33
7.7	Regression	33
8	Bibliography	35
	Bibliography	37

Physical models, thermodynamics, heat and mass transfer, and validation of the thermopack-based CO₂ release / dry-ice add-on to HydDown.

Version: 1.0
Date: September 2025
Author: Anders Andreasen
Branch: co2-release-hem

This technical reference documents the CO₂ **release and dry-ice** add-on developed on the `co2-release-hem` branch of HydDown [1]. It covers the additional physical models, the thermopack thermodynamic backend and its hand-overs with CoolProp, the homogeneous-equilibrium (HEM) and non-equilibrium (HNE) discharge models, the in-vessel and atmospheric dry-ice estimation, the heat-and mass-transfer treatment, and the validation against the CARDICE Joint Industry Project experiments (Ineris 2 m³ sphere).

It is a companion to the base HydDown *Manual* (`Manual.md`); only the CO₂ extension is described here. The document follows the structure of the ThermoCalc technical reference.

INTRODUCTION

1.1 Motivation

In carbon capture and storage (CCS) with ship or truck transport, CO₂ is handled as a pressurised, liquefied fluid at vapour/liquid equilibrium. An accidental leak - at an onshore tank, on a ship, or during transfer - depressurises the inventory towards atmospheric pressure. Because the atmospheric pressure of CO₂ (1 atm) lies **below the triple point** ($P_{tp} = 5.18$ bar, $T_{tp} = -56.6$ °C), the expanding stream necessarily passes into the **vapour + solid** region: it forms **dry ice** (“CO₂ snow”) both downstream of the leak and, for a vapour-space release, as a large retained bank inside the vessel. Dry-ice formation drives long shutdowns (sublimation is slow) and severe low wall temperatures that govern material selection.

The base HydDown code [1] uses CoolProp [2] as its thermodynamic backend. CoolProp’s CO₂ equation of state [3] is **not valid below the triple point**, so it can neither integrate a homogeneous-equilibrium (HEM) discharge isentrope through a sub-triple throat nor evaluate the atmospheric dry-ice end state. The CO₂ release add-on therefore introduces a **thermopack** [4] backend with a solid CO₂ model, used only for the parts of the calculation that reach below the triple point, while CoolProp continues to drive the vessel thermodynamics above it.

1.2 Scope

The add-on is tailored to **pure CO₂** and to two release geometries, matching the CARDICE experiments (*Experimental Basis (CARDICE)*):

- **Gas (vapour-space) release** - a hole in the vapour space; saturated-vapour stagnation. The liquid stays in the vessel, cools to the triple point, freezes to a large retained dry-ice bank, and the vessel finally blows down on the sublimation line. This is the classic “hundreds of kilograms of snow in the vessel” case.
- **Liquid (bottom) release** - a hole in the liquid space; saturated-liquid stagnation. The liquid drains (and boils off) and is essentially gone *before* the vessel reaches the triple point, so **no significant in-vessel dry ice** forms; the dry ice appears downstream of the orifice instead.

Both cases require the two-phase (non-equilibrium) vessel model (`calculation.non_equilibrium: true`, `calculation.type: energybalance`), a liquid fill (`vessel.liquid_level`), and the new top-level `release: block`.

1.3 What the add-on provides

Table 1.1: Physical models added on co2-release-hem

Area	Model
Thermodynamics	thermopack tc-PR [5] with a solid CO ₂ model; a self-consistent triple point; CoolProp ↔ thermopack hand-overs (<i>Thermodynamics and Hand-overs</i>).
Discharge	homogeneous-equilibrium (HEM) rate with a solid-aware throat; an optional non-equilibrium (HNE) liquid boost for delayed flashing (<i>Discharge Model</i>).
Dry ice	atmospheric (downstream) vapour/solid split by an isenthalpic lever; an opt-in in-vessel solid model (single-zone and two-zone plateau/descent) (<i>Dry-Ice Estimation</i>).
Heat transfer	a two-node (gas-contact / wetted) wall; a natural-convection gas-wall HTC valid below the triple point; superheated gas and split wall initialisation (<i>Heat and Mass Transfer</i>).
Mass / energy	separate gas- and liquid-zone balances with phase-attributed outflow enthalpy for a liquid- vs vapour-space release (<i>Heat and Mass Transfer</i>).
Validation	the full CARDICE pure-CO ₂ test set (tests 5-10) against 1 Hz Ineris data (<i>Validation</i>).

1.4 Architecture

All thermopack work is encapsulated in a standalone module, `src/hyddown/co2_release.py` (class `CO2ReleaseModel`); the main `HydDown` class in `hdclass.py` only calls thin glue, so thermopack is never imported for a non-release run. The vessel state above the triple point is tracked by the existing CoolProp non-equilibrium model (NEM); once the tank approaches the triple point the state is handed to the thermopack solid-in-vessel model.

A minimal input looks like:

```
vessel:      {length: 0.0, diameter: 1.58, type: Hemispherical,
              liquid_level: 0.62, thickness: 0.072, ...}
initial:     {temperature: 253.15, pressure: 2000000, fluid: "CO2",
              gas_temperature: 256.2}                # superheated gas zone
↔(optional)
calculation: {type: energybalance, non_equilibrium: true,
              time_step: 5, end_time: 13000, h_gas_liquid: 2.0}
valve:       {type: none, flow: discharge, back_pressure: 101325.}
release:     # <-- the CO2 add-on
              # or "gas"
              type: liquid
              diameter: 0.00266
              discharge_coef: 0.68
              liquid_nonequilibrium: 0.267           # HNE boost (liquid only)
              eos: tcPR
              solid_in_vessel: true
              back_pressure: 101325.
              atm_pressure: 101325.
```

(continues on next page)

(continued from previous page)

```
heat_transfer: {type: specified_h, temp_ambient: 283.15,  
                h_outer: 0.4, h_inner: calc}
```

Worked examples ship in `src/hyddown/examples/` (`co2_release.yml`, `co2_gas_release.yml`, `co2_solid_in_vessel.yml`) and the validation cases in `validation/CARDICE_*.yml`.

EXPERIMENTAL BASIS (CARDICE)

The add-on is validated against the **CARDICE** (CARbon Dioxide ICE) Joint Industry Project, in which Ineris performed pilot-scale CO₂ blowdown experiments in a 2 m³ sphere. Two papers document the work: the GHGT-15 paper [6], which gives the test matrix and the Vessfire comparison, and the IJGGC setup paper [7], which documents the rig, instrumentation and calibration in detail. The open 1 Hz dataset is on Zenodo [8].

2.1 The vessel

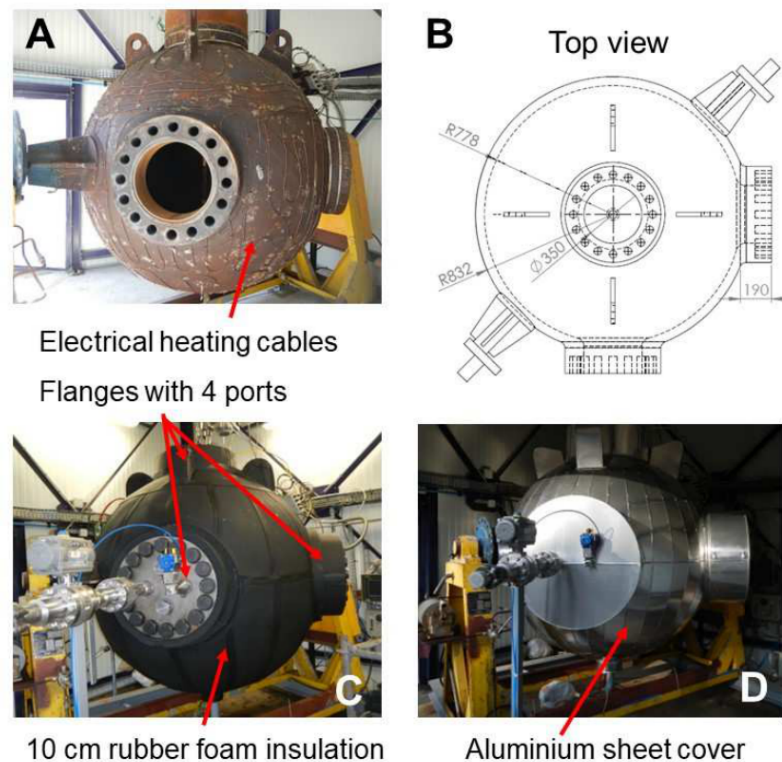


Figure 1: The 2m³ steel vessel. (A) bare, (B) Top view drawing, (C) with the 10 cm rubber foam insulation, (D) with aluminium sheet cover

Fig. 2.1: The 2 m³ steel sphere: (A) bare, (B) top-view drawing, (C) with flanges and insulation, (D) with the aluminium cover.

From [7].

The vessel characteristics reported by [7] are summarised in Table 2.1.

Table 2.1: Vessel characteristics

Property	Value
Geometry	sphere, two welded hemispheres
Internal volume	$2.030 \pm 0.005 \text{ m}^3$ (geometric)
Inner radius / diameter	778 mm / 1556 mm
Mean wall thickness	$54 \pm 5 \text{ mm}$
Total vessel mass	4.80 t (steel shell $\approx 4 \text{ t} + 3 \text{ flanges}$)
Steel grade	P355GH (1.0473)
Heat capacity c_p	450 J/(kg·K)
Thermal conductivity λ	44 W/(m·K)
Density	7800 kg/m ³
Design	200 bara / $-60 \text{ }^\circ\text{C}$

In HydDown the sphere is modelled as two hemispherical heads with zero cylinder length (vessel.type: Hemispherical, length: 0, diameter: 1.58). The wall thickness is set to 0.072 m rather than the physical 54 mm so that the shell-corrected wall mass matches the measured 4.80 t total (the extra accounts for the three heavy flanges/nozzles, which HydDown does not model separately).

2.2 Insulation and external heat transfer

The sphere is insulated with 10 cm of rubber foam ($\lambda_{\text{fo}} = 0.034 \text{ W/(m·K)}$) over an outer area of 11 m^2 . [7] reports the external heat transfer in two ways:

Theoretical (steady conduction through the foam):

$$h_{\text{theo}} = \frac{\lambda_{\text{fo}} A_{\text{out}}}{L_{\text{fo}}} = \frac{0.034 \times 11}{0.10} = 3.75 \text{ W/K} \quad (2.1)$$

Measured (a cooling calorimetry test, their Fig. 2): the insulated vessel was warmed and allowed to cool towards the $2 \text{ }^\circ\text{C}$ ambient. The contents dropped $\Delta T = 8 \text{ }^\circ\text{C}$ over $\Delta t = 45 \text{ h}$. The heat-loss rate is inferred from the vessel thermal mass C (dominated by the 4 t of steel, $4000 \times 450 = 1.8 \text{ MJ/K}$, giving $\approx 89 \text{ W}$; $\approx 107 \text{ W}$ including the contents):

$$\dot{Q} = C \frac{\Delta T}{\Delta t} \approx 107 \text{ W}, \quad h = \frac{\dot{Q}}{\Delta T} = \frac{107}{25} = 4.3 \text{ W/K} \quad (2.2)$$

where $\overline{\Delta T} \approx 25 \text{ }^\circ\text{C}$ is the mean inside-outside gap over the run. Note that this h is a **whole-vessel conductance in W/K**, not a per-area coefficient; per unit outer area it is $4.3/11 = 0.39 \text{ W/(m}^2\cdot\text{K)}$, which is the value used as `heat_transfer.h_outer` in the models. The measured 4.3 W/K exceeds the pure-foam 3.75 W/K by $\approx 15 \%$ because the real heat paths also include the flanges, the release pipe, and the thermocouple wells ($8 \text{ mm} \times 250 \text{ mm}$ pipes protruding through the insulation) - thermal bridges the conduction estimate ignores.

For the several-hour CO_2 blowdowns this $\sim 100 \text{ W}$ ingress is small against the latent (flash-ing/sublimation) loads, so the blowdown is effectively near-adiabatic.

2.3 Instrumentation

The measured quantities are:

- absolute **pressure** inside the sphere and in the release pipe;

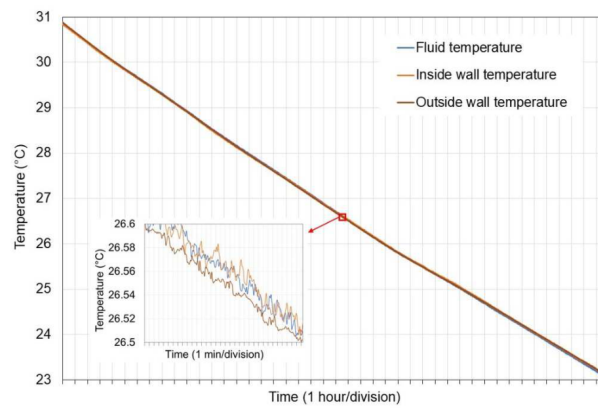


Figure 2: Temperature curves recorded during the cooling of the insulated vessel from 31°C to ambient (outside temperature = 2°C)

Fig. 2.2: Cooling curves of the insulated vessel used to obtain the measured external heat-transfer conductance.

From [7].

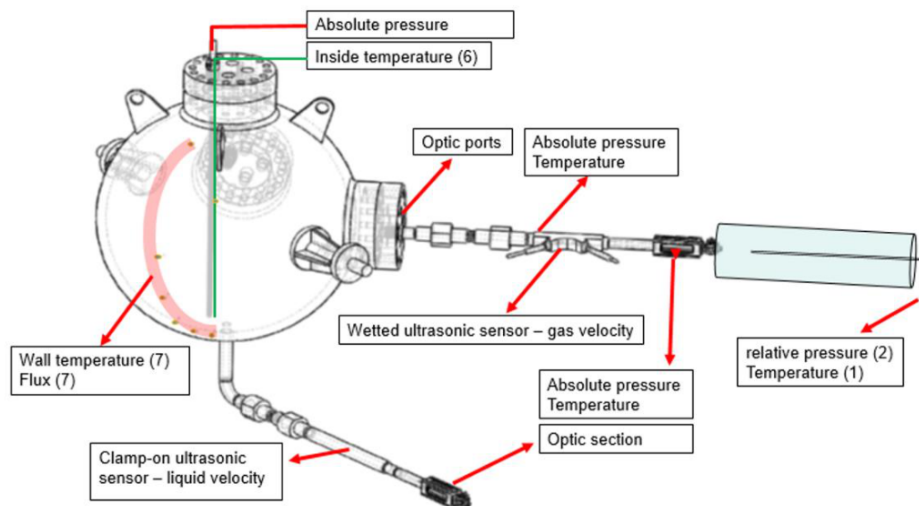


Fig. 2. Overall instrumentation of the experimental set-up.

Fig. 2.3: Overall instrumentation of the set-up.

From [6].

- **fluid temperature** at 6 heights inside the sphere (T_c in 1..6) and in the pipe;
- inner and outer **wall temperature** at 7 heights ($T_{c1}..T_{c7}$ / $T_{cF1}..T_{cF7}$);
- **mass flow rate** through the release pipe;
- the **mass of the vessel** (fluid + solids) via 4 load cells (Mettler-Toledo 0745A, 2200 kg each, ± 100 g; total-mass uncertainty ± 0.4 t on 4 t; flow-rate uncertainty ± 5 %).

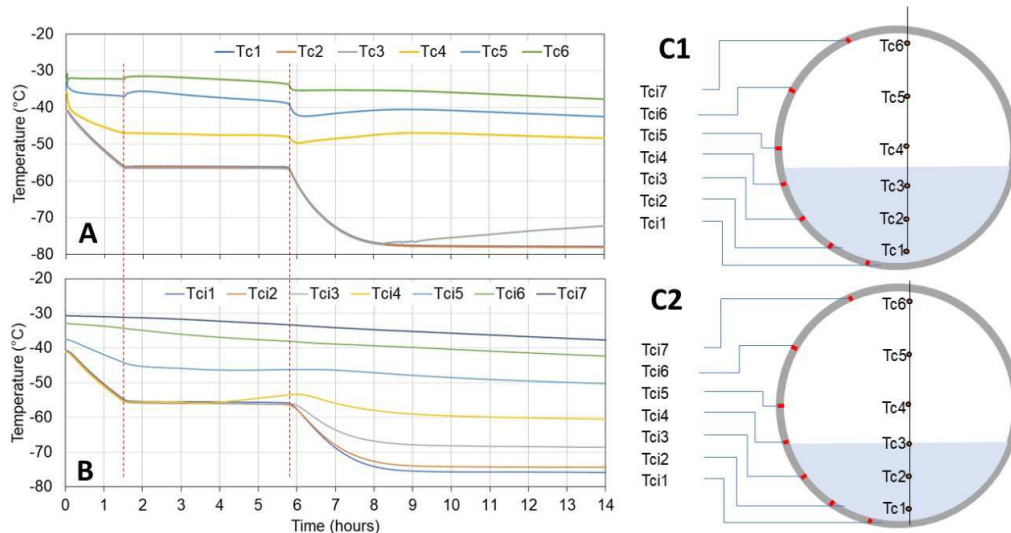


Figure 16: Temperature profile of the fluid. (A) inside the vessel (T_{cx}) and (B) in the inner wall surface (T_{cix}) during the release. On the right side, drawings of the vessel showing (C1) the location of the gas/liquid interface at $t=0$ (C1) and (C2) the

Fig. 2.4: Example measured temperatures during a blowdown: (A) fluid at 6 heights inside the vessel and (B) the wall 5 mm from the inner face, showing the stable vertical stratification of the vapour phase. From [7].

Two independent flow measurements were used: direct **weighing** (load cells) and an **ultrasonic** velocity (inline on the gas pipe, clamp-on on the liquid pipe) combined with the computed density. The load-cell mass, corroborated against the measured temperature and pressure, is the reference used here for the initial inventory and the discharge-rate calibration. A camera and a katharometer were also fitted.

The release pipe is 40 mm inner diameter, ~ 2 m long, terminated by a **calibrated orifice** connected to the vessel middle (vapour release) or bottom (liquid release). The paper gives one pure- CO_2 orifice example of **4 mm** (range 1 mm to full bore); per-test sizes are not reported, so they are inferred here from the measured discharge (*Discharge Model*).

2.4 Test matrix

The pure- CO_2 tests are 5-10; tests 1-4 are methane / methane- CO_2 mixtures and are out of scope. For the low-temperature tests the sphere was filled with liquid to about the middle. The matrix is reported by [6].

Table 2.2: CARDICE pure-CO₂ test matrix

Test	Composition	P_0 [bar]	T_0 [°C]	Release	HydDown file
5	100 % CO ₂	20	−20	Gas	CARDICE_test5.yml
6	100 % CO ₂	20	−20	Liquid	CARDICE_batch6.yml
7	100 % CO ₂	15	−29	Gas	CARDICE_test7.yml
8	100 % CO ₂	15	−29	Liquid	CARDICE_test8.yml
9	100 % CO ₂	10	−40	Gas	CARDICE_test9.yml
10	100 % CO ₂	10	−40	Liquid	CARDICE_test10.yml

Gas releases produced large in-vessel dry-ice banks (over 450 kg reported), as in Fig. 2.5; liquid releases produced essentially none inside the vessel.

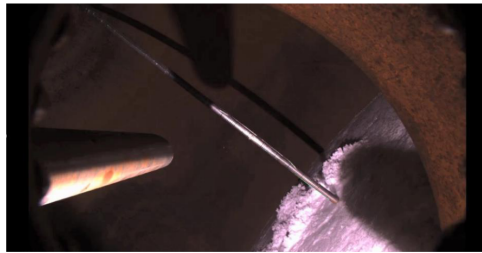
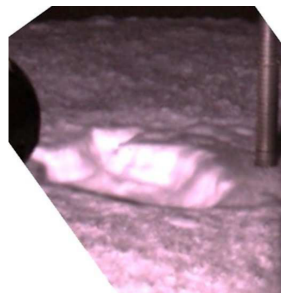


Fig. 3. Solid CO₂ formed inside the sphere during a gas release test. The picture is tilted, the vertical is indicated by the metal rod in contact with dry ice.

Fig. 2.5: Solid CO₂ formed inside the sphere during a **gas** release test (the picture is tilted; the metal rod indicates the vertical).

From [6].



596 Figure 19: Close-up showing the dry ice foam-like structure. Image extracted from a
597 video taken several hours after complete vaporization of the liquid phase
598
599

Fig. 2.6: Close-up of the dry-ice “foam-like” structure (floating solid with vapour bubbles).

From [7].

2.5 Raw-data alignment

The 1 Hz files include a pre-release period; the true blowdown start t_0 is detected from the onset of the inventory-mass decline (and the pressure drop) and all comparisons are time-shifted to it (e.g. $t_0 = 438$ s for cardice-06). Folder `cardice-0N` corresponds to test N .

THERMODYNAMICS AND HAND-OVERS

This chapter describes the thermopack backend, the CO₂ phase regimes, the self-consistent triple point, and the hand-overs between CoolProp and thermopack. All of it is implemented in `co2_release.py` (class `CO2ReleaseModel`).

3.1 Why a second backend

The atmospheric pressure of CO₂ lies below its triple point, so a release crosses into the **vapour + solid** region. CoolProp's reference CO₂ equation of state [3] is not defined there, so it cannot:

- integrate the HEM discharge isentrope through a throat that drops below 5.18 bar, nor
- evaluate the atmospheric (1 atm) dry-ice end state.

thermopack [4] is used for exactly these sub-triple tasks. The translated-consistent Peng-Robinson EoS (tc-PR) [5] is initialised with a solid CO₂ model:

```
self.eos = tcPR("CO2")
self.eos.init_solid("CO2")
```

Its solid-aware `two_phase_psflash` returns a vapour+solid state (phase index 7, with the **solid mole fraction in the betaL slot**). Keeping every sub-triple property inside thermopack means a single, consistent reference basis - the code never mixes CoolProp and thermopack enthalpies/entropies (which differ by a large, fixed reference offset). eos: tcPR is the only supported option because the solid model is calibrated against it.

3.2 Phase regimes

A CO₂ release passes through three regimes:

1. **Above the triple point** - liquid + vapour equilibrium. The vessel is tracked by the existing CoolProp non-equilibrium model (NEM); discharge stagnation states are saturated liquid or vapour.
2. **At the triple point** - solid + liquid + vapour coexist. By Gibbs' phase rule a pure three-phase point is invariant: T and P are pinned while the leak and heat only shift the phase *amounts* (the freezing "lever"). This is the pressure **plateau** seen in gas-release blowdowns.
3. **Below the triple point** - vapour + solid on the **sublimation line**. Once the liquid is exhausted the state rides $T_{\text{sub}}(P)$ down to the back pressure (frost point $\approx 194.14 \text{ K} / -79^\circ \text{C}$ at 1 atm).

3.3 The self-consistent triple point

The literature triple point is $P_{\text{tp}} = 5.18$ bar, $T_{\text{tp}} = 216.59$ K, but tc-PR has its **own** self-consistent triple point, which the code locates so that the solid and fluid models are thermodynamically consistent (no spurious latent-heat jumps). The triple point is where the solid and liquid Gibbs energies are equal along the fluid saturation line (on which $g_{\text{liquid}} = g_{\text{vapour}}$ already, for a pure component):

$$g_{\text{solid}}(T, P_{\text{sat}}(T)) = g_{\text{liquid}}(T, P_{\text{sat}}(T)), \quad g = h - Ts \quad (3.1)$$

Solving (`_init_triple_point`) gives $T_{\text{tp}}^{\text{EoS}} = 216.59$ K, $P_{\text{tp}}^{\text{EoS}} = 5.255$ bar. The pure-phase specific volumes, enthalpies and internal energies of solid, liquid and vapour at this point are stored as the vertices of the three-phase lever.

3.4 Hand-overs

Two distinct hand-overs occur; understanding them is key to the model.

3.4.1 Vessel state: CoolProp → thermopack (at the triple point)

Above the triple point the vessel inventory is integrated by the CoolProp NEM. Once the tank pressure reaches $P \leq 1.05 P_{\text{tp}}^{\text{EoS}}$ and the opt-in `release.solid_in_vessel` is set, the state is handed to the thermopack solid-in-vessel model (`hdclass.py`). The NEM's two zones - a warm (superheated) gas zone and a liquid/solid zone - are re-based into thermopack's energy basis (masses are basis-free), and a single-zone ($M_{\text{vessel}}, U_{\text{vessel}}$) state is also seeded for the sublimation descent. From then on the below-triple model (*Dry-Ice Estimation*) advances the vessel.

3.4.2 Discharge: always thermopack

The **discharge rate** is computed by the thermopack HEM at *all* pressures - above the triple point too - so there is no discontinuity in the mass rate at the triple-point crossing. Only the *vessel state* hands over; the discharge model does not. A fine-resolution check across the triple point shows the rate is smooth and matches a choked-orifice value in both regimes (*Discharge Model*).

Note

The gas *temperature* is not C^1 -continuous across the vessel hand-over: the below-triple descent cools the residual gas a few degrees more steeply than the NEM just above (a $\sim 3^\circ\text{C}$ kink for liquid-release cases). This is small compared with the 0-D stratification gap (*Heat and Mass Transfer*) and is a known, deferred item.

3.5 Precomputed property tables

The below-triple two-zone model steps the gas zone many times with per-step root-finds; live thermopack flashes would be far too slow. `_init_gas_table` therefore precomputes (once, ~ 1 s):

- a **1-D table** of superheated-gas properties (u, ρ, h , HEM flux G) versus T at the triple-point pressure - used on the plateau, where P is fixed;
- a **2-D table** $(T, P) \rightarrow (\rho, u, h)$ for the descent, where P floats below the triple point;
- the **sublimation line** $T_{\text{sub}}(P)$ and the saturated-vapour / solid internal energies along it.

The descent leak rate itself is evaluated live per step (there are far fewer descent steps than table nodes). Interpolants (`gas_T_from_u`, `_gas2d`, `_T_sub_of_P`, ...) provide the state during the integration.

DISCHARGE MODEL

The release rate is computed by a homogeneous-equilibrium (HEM) model with a solid-aware throat, plus an optional non-equilibrium (HNE) boost for a flashing liquid. Implemented in `co2_release.py`.

4.1 Stagnation state

The upstream (stagnation) state feeding the hole follows `release.type` and, once the liquid is exhausted, the phase flip performed in the mass balance (`release_phase liquid → gas`):

- **liquid release** - saturated liquid at the tank pressure P_0 (`stagnation(P, "liquid")`);
- **gas release** - saturated vapour at P_0 (`stagnation(P, "gas")`);
- **below the triple point** - an explicit vapour state on the sublimation line (`gas_leak_rate(T, P)`), where a saturated stagnation is undefined.

Each returns the mass-specific stagnation enthalpy h_0 , molar entropy s_0 , density ρ_0 and temperature T_0 .

4.2 HEM mass flux

The flow is assumed homogeneous (no slip) and isentropic from the stagnation state to the throat. Along the isentrope $s = s_0$, the mass flux at throat pressure P is

$$G(P) = \rho(P) \sqrt{2(h_0 - h(P))}, \quad s(P) = s_0, \quad (4.1)$$

and the discharge maximises G over the throat pressure. The maximiser is the **choked throat**; if G keeps rising down to the back pressure the flow is unchoked and the throat sits at P_{back} :

$$\dot{m} = C_d A G_{\text{max}}, \quad G_{\text{max}} = \max_{P_{\text{back}} \leq P \leq P_0} G(P). \quad (4.2)$$

The isentrope properties come from thermopack's **solid-aware** `two_phase_psflash(_iso_props)`: below the triple pressure the throat state is a vapour + dry-ice mixture, and its density/enthalpy carry the solid fraction - the “dry-ice lever at the throat”. Consequently a liquid or gas stream that expands below the triple point correctly forms solid at the orifice.

Note

The flux curve $G(P)$ can carry **two local maxima** across the triple-point kink (one on the vapour+liquid branch above 5.18 bar, one on the vapour+solid branch below it). A bare bounded search can lock onto the wrong one, so `hem_rate_from_stagnation` first locates the global maximum on a coarse 40-point grid and then refines locally (`scipy.optimize.minimize_scalar, bounded`).

The returned dictionary carries $\dot{m}$, G , the throat pressure/temperature, the throat solid fraction, a choked flag, and the stagnation state.

Implementation: `co2_release.py:hem_rate()` / `hem_rate_from_stagnation()`.

4.2.1 Verification against theory

At a representative gas point (saturated vapour, 12 bar, 6 mm, $C_d = 0.68$) the thermopack HEM, a CoolProp HEOS HEM, and a textbook **choked ideal-gas orifice** agree to within a few percent:

Method	G [kg/m ² s]	$\dot{m}$ [kg/s]
thermopack tc-PR HEM	3861	0.0742
CoolProp HEOS HEM	3878	0.0746
choked ideal-gas orifice	3755	0.0722

So the HEM flux is physically correct; it is the *feed* (which phase, and the effective discharge coefficient) that carries the modelling choices.

4.3 Non-equilibrium liquid discharge (HNE)

A **saturated** liquid discharged through a **short, sharp orifice** has too little residence time to flash to equilibrium: the metastable liquid persists past its saturation pressure and discharges *above* the equilibrium-HEM rate, toward the frozen all-liquid limit [9, 10]. The equilibrium HEM (instantaneous flashing) is the low bound; the frozen incompressible (Bernoulli) flux is the high bound:

$$G_{\text{frozen}} = \sqrt{2 \rho_L (P_0 - P_{\text{back}})}. \quad (4.3)$$

The add-on interpolates between them with a single non-equilibrium factor $N \in [0, 1]$ (`release.liquid_nonequilibrium`):

$$G_{\text{liq}} = \sqrt{(1 - N) G_{\text{HEM}}^2 + N G_{\text{frozen}}^2} \quad (4.4)$$

so $N = 0$ recovers equilibrium HEM and $N = 1$ the frozen liquid. The blend is applied only to a **liquid** stagnation; a gas discharge (single phase, nothing to flash) is untouched. The default is $N = 0$, so existing runs are unchanged.

This is a deliberately simple metastable interpolation. A fuller non-equilibrium model for CO₂ releases is the Vianna/Lopes HNM [11], which adds a bubble-nucleation relaxation factor across three isentropic segments (liquid → saturation → triple point → atmosphere). That model is derived and validated for **initially sub-cooled / dense-phase liquid at high pressure** (49-159 bar) and explicitly excludes gas-space releases; its dominant incompressible term vanishes at the saturated bubble point. It is therefore not directly applicable to the CARDICE saturated, low-pressure (10-20 bar) cases, where the single-factor blend (4.4) is sufficient. The Leung ω -method [12] provides an alternative subcooled/flashing correlation.

Implementation: `co2_release.py:hem_rate()` (liquid branch).

4.4 Orifice and discharge coefficient

Because the per-test orifice size is not reported, it is inferred from the measured discharge with a single physical coefficient $C_d = 0.68$. The **gas** discharge is reliable HEM (it equals a choked orifice), so for a

gas release, and for the *gas tail* of a liquid release, the measured gas rate fixes the orifice. The **liquid** rate of that same test is then matched with the HNE factor N . The inferred sizes:

Table 4.1: Estimated orifices (single $C_d = 0.68$)

Test	Phase	P_0 [bar]	Orifice	N	Inferred from
5	gas	20	3.44 mm	–	gas rate
6	liquid	20	2.66 mm	0.267	gas tail + liquid rate
7	gas	15	4.38 mm	–	gas rate
8	liquid	15	4.70 mm	0.195	gas tail + liquid rate
9	gas	10	4.45 mm	–	gas rate
10	liquid	10	4.09 mm	0.0	liquid rate

The metastable boost N decreases with pressure (0.267 at 20 bar, 0.195 at 15 bar, 0 at 10 bar); at 10 bar the equilibrium HEM liquid flux already meets the data, so test 10's orifice is taken from the liquid rate directly (the gas-inferred value would over-drain it). The inferred sizes (2.66-4.70 mm) sit inside the papers' 1 mm-to-full-bore range and near the one documented 4 mm example. A plausible physical reading is that tests 7-10 share a nominal 4 mm orifice and the spread comes from using one C_d ; splitting C_d by phase (gas ~ 0.84 -0.9, liquid ~ 0.62 -0.65) would collapse them toward 4 mm - a possible future refinement.

4.4.1 Why the single- C_d + HNE convention

If the orifice is instead inferred from the *liquid* rate assuming equilibrium HEM, it comes out too large (HEM under-predicts the flashing liquid), and that oversized orifice then makes the *gas tail* $\sim 2\times$ too fast. Inferring the orifice from the reliable gas discharge and giving the liquid the HNE boost removes this inconsistency and lets one C_d fit both phases (e.g. test 8: liquid 0.311 vs 0.317 kg/s, gas tail 0.030 vs 0.030 kg/s, both matched).

DRY-ICE ESTIMATION

Two dry-ice quantities are estimated: the **atmospheric** (downstream) vapour/solid split of the released stream, which is always computed, and the **in-vessel** solid that accumulates when a gas release drags the vessel below the triple point, which is an opt-in model. All of it is in `co2_release.py` (atmospheric) and the two-zone / solid-regime steppers dispatched from `hdclass.py`.

5.1 Atmospheric (downstream) dry ice

A free jet does no external work, so the released stream conserves its stagnation enthalpy as it flashes to atmospheric pressure. `thermopack`'s `two_phase_phflash` is not solid-aware, so the vapour/solid split is computed with an explicit **isenthalpic lever** against pre-computed 1-atm endpoint enthalpies (the pure-vapour $h_{\text{gas}}^{\text{atm}}$ and pure-solid $h_{\text{solid}}^{\text{atm}}$ at the frost point):

$$\beta_{\text{gas}} = \frac{h_0 - h_{\text{solid}}^{\text{atm}}}{h_{\text{gas}}^{\text{atm}} - h_{\text{solid}}^{\text{atm}}}, \quad x_{\text{solid}} = 1 - \text{clip}(\beta_{\text{gas}}, 0, 1) \quad (5.1)$$

If $\beta_{\text{gas}} \geq 1$ the stream is superheated (no dry ice) and a plain solid-free flash gives its temperature; otherwise the stream is a vapour + dry-ice mixture at the **frost point** $T_{\text{frost}} \approx 194.14 \text{ K}$ (-79°C), obtained directly from a `psflash` whose entropy brackets the gas/solid values (the gas $\leftrightarrow$ solid Gibbs-equality point). The endpoints depend only on the back pressure and are computed once (`_init_atm_endpoints`).

An **isentropic** variant (`atm_split_isentropic`) uses the solid-aware `psflash` directly and gives the available-work / blast-energy bound.

For a liquid release the discharged stream is typically $\sim 50\%$ dry ice by mass at 1 atm - the dry ice a liquid leak produces appears **here**, downstream of the orifice, not inside the vessel.

Implementation: `co2_release.py:atm_split()` / `atm_split_isentropic()`.

5.2 In-vessel dry ice (opt-in)

When a **gas** release keeps the liquid in the vessel, the liquid cools to the triple point, freezes, and a large dry-ice bank is retained; the vessel finally blows down on the sublimation line. This is enabled with `release.solid_in_vessel: true`. Once the tank reaches the triple point (*Thermodynamics and Hand-overs*) the vessel is advanced by a `thermopack` model with two regimes: a **triple-point plateau** while liquid remains, and a **sublimation descent** afterwards. Two implementations exist; the default (used for the CARDICE gas cases) is the two-zone model.

5.2.1 Triple-point plateau (three-phase lever)

At the invariant triple point the temperature and pressure are pinned and the specific volumes and internal energies of the three pure phases ($v_g, v_l, v_s, u_g, u_l, u_s$) are constants. The phase masses then follow from conservation of total mass, volume and internal energy:

$$\begin{aligned} m_g + m_l + m_s &= M \\ m_g v_g + m_l v_l + m_s v_s &= V \\ m_g u_g + m_l u_l + m_s u_s &= U \end{aligned} \quad (5.2)$$

As the leak removes mass and (small) heat enters, M and U change and the lever redistributes the phases - the liquid steadily converts to solid at fixed T, P . This produces the observed **pressure plateau** at the triple point.

In the **two-zone** plateau (`two_zone_plateau_step`) the gas is carried as a separate warm (superheated) zone that keeps its own temperature - heated by the wall and only weakly coupled to the cold liquid/solid - so the model reproduces the measured gas superheat *and* the freezing lever simultaneously. The gas fills the vapour volume at the triple-point pressure.

5.2.2 Sublimation descent

Once the liquid is exhausted the state rides the sublimation line. In the two-zone descent (`two_zone_descent_step`) a warm gas zone leaks and depressurises while the dry-ice bank sublimates only enough to (a) cool itself as $T_{\text{sub}}(P)$ falls and (b) absorb the gas $\rightarrow$ solid interphase heat $\dot{Q}_{gs} = UA_{gs}(T_g - T_s)$:

$$\dot{m}_{\text{sub}} = \underbrace{\frac{m_s c_{p,s} (T_s^- - T_s)}{L_{\text{sub}}}}_{\text{self-cooling}} + \underbrace{\frac{\dot{Q}_{gs} \Delta t}{L_{\text{sub}}}}_{\text{interphase}} \quad (5.3)$$

The interphase conductance UA_{gs} (from `release.solid_h_gas_solid`) is the single lever that trades gas superheat against retained solid: with $UA_{gs} = 0$ the dry ice is adiabatic and mostly retained. The gas energy balance carries the wall heat, the outflow enthalpy and the sublimed-vapour enthalpy; the solid mass is decremented by $\dot{m}_{\text{sub}}$.

5.2.3 Two-zone parameters

Parameter	Meaning
<code>solid_h_gas_wall</code>	gas $\leftrightarrow$ wall HTC below the triple point; a number, or "calc" for a natural-convection estimate (<i>Heat and Mass Transfer</i>).
<code>solid_h_gas_liquid</code>	plateau gas $\leftrightarrow$ liquid/solid interphase HTC (small - keeps the gas warm; ≈ 0 for a flat liquid surface).
<code>solid_h_gas_solid</code>	descent gas $\leftrightarrow$ dry-ice interphase conductance (trades gas superheat vs retained solid; $\approx 3 \text{ W/m}^2\text{K}$ calibrated).
<code>solid_gas_wall_frac</code>	fraction of the inner wall in gas contact (the rest is wetted).
<code>solid_h_inner</code>	wall $\leftrightarrow$ solid/liquid HTC for the wetted wall node (<i>Heat and Mass Transfer</i>).

5.3 Behaviour

The model reproduces the central experimental contrast:

- **Gas releases** (tests 5/7/9) retain a large in-vessel dry-ice bank ($\sim 270/310/400$ kg modelled vs $\sim 217/298/412$ kg measured) with a multi-hour triple-point plateau.
- **Liquid releases** (tests 6/8/10) retain essentially none ($\sim 4\text{-}5$ kg vs ~ 0): the liquid drains and boils out **before** the vessel reaches the triple point, so it never freezes in place. The dry ice appears downstream instead ((5.1)).

See *Validation* for the full comparison.

HEAT AND MASS TRANSFER

The two-phase (non-equilibrium) vessel carries **separate gas and liquid zones** with their own temperatures, and a **two-node wall** (a gas-contact node and a liquid/solid-contact “wetted” node). This chapter documents the wall model above and below the triple point, the natural-convection coefficients, the initialisation, and the zone mass/energy balances. Implemented in `hdclass.py` (with correlations in `transport.py` and the below-triple wall in the two-zone steppers).

6.1 Two-node wall above the triple point (NEM)

Above the triple point the CoolProp non-equilibrium model resolves the wall into an **unwetted (gas-contact)** node T_{wall} and a **wetted (liquid-contact)** node T_{wall}^w , exchanging with the respective fluid zone:

$$\begin{aligned}\dot{Q}_{\text{inner}} &= h_{\text{in}} A_{\text{unwetted}} (T_{\text{wall}} - T_g) \\ \dot{Q}_{\text{inner}}^w &= h_{\text{in}}^w A_{\text{wetted}} (T_{\text{wall}}^w - T_l)\end{aligned}\tag{6.1}$$

with the wetted/unwetted areas from the vessel geometry at the current liquid level. The inner coefficients h_{in} follow natural-convection correlations (`h_inner: calc`); the gas side uses the **gas** temperature in the NEM. The external side uses `h_outer` (the Jamois-measured $0.39 \approx 0.4 \text{ W/m}^2\text{K}$, *Experimental Basis (CARDICE)*).

6.2 Natural convection

The natural-convection HTC uses the Geankoplis [13] Nusselt correlation with the Rayleigh number $\text{Ra} = \text{Gr Pr}$:

$$\text{Gr} = \frac{g \beta |T_{\text{wall}} - T_{\text{fluid}}| L^3}{\nu^2}, \quad \text{Nu} = \begin{cases} 0.13 \text{Ra}^{1/3} & \text{Ra} \geq 10^9 \\ 0.59 \text{Ra}^{1/4} & 10^4 < \text{Ra} < 10^9 \\ 1.36 \text{Ra}^{1/5} & \text{otherwise} \end{cases} \quad h = \frac{\text{Nu } k}{L} \tag{6.2}$$

with L the vessel diameter and fluid properties (β, ν, k, Pr) from CoolProp.

6.3 Below the triple point (two-zone wall)

Below the triple point the two-zone model carries the wall as two nodes.

Gas-contact wall (T_{wall}) exchanges with the gas and the ambient (the liquid/solid zone is adiabatic to it):

$$m_w c_{p,w} \frac{dT_{\text{wall}}}{dt} = \dot{Q}_{\text{out}} - \dot{Q}_{wg}, \quad \dot{Q}_{wg} = h_{gw} A_g (T_{\text{wall}} - T_g), \quad \dot{Q}_{\text{out}} = h_{\text{out}} A_{\text{out}} (T_{\text{amb}} - T_{\text{wall}}) \tag{6.3}$$

where $A_g = A_{\text{inner}} \cdot \text{solid_gas_wall_frac}$.

Wetted (solid/liquid-contact) wall (T_{wall}^w , `_wetted_wall_step`) relaxes toward the cold phase - the triple point on the plateau, the sublimation line on the descent - over the wetted wall fraction, using `solid_h_inner` as the wall $\leftrightarrow$ solid coefficient:

$$m_w^w c_{p,w} \frac{dT_{\text{wall}}^w}{dt} = \dot{Q}_{\text{out}}^w - \dot{Q}_{ws}, \quad \dot{Q}_{ws} = h_{\text{si}} A_{\text{wet}} (T_{\text{wall}}^w - T_{\text{cold}}) \quad (6.4)$$

Wetted-wall hand-over. The wetted node is *seeded at the triple-point hand-over from the above-triple wetted wall* (which has cooled with the liquid), so the cold liquid-contact wall keeps tracking the cold phase instead of jumping back to the warm gas wall. Before this fix the reported wetted wall collapsed to the gas wall at the triple point (a $\sim 20^\circ\text{C}$ jump). The reciprocal wall $\rightarrow$ solid heat is intentionally **not** removed from the (adiabatic) solid zone, matching the two-zone simplification and leaving the calibrated dry-ice mass unchanged. With `solid_h_inner` = 150 W/m²K the wetted wall reaches $\approx -75^\circ\text{C}$ (matching the measured bottom sensors); a sensitivity on test 7 gives $20 \rightarrow -59$, $60 \rightarrow -70$, $150 \rightarrow -74$, $400 \rightarrow -75.5^\circ\text{C}$.

6.4 Natural-convection gas-wall HTC (“calc”)

`release.solid_h_gas_wall` accepts a fixed number or the string “calc”. With “calc” the below-triple gas $\leftrightarrow$ wall coefficient is the same natural-convection correlation (6.2), evaluated for CO₂ **vapour** via CoolProp’s T|gas phase specification - which is valid for the gas root below the triple point (verified down to ~ 1 bar). At plateau conditions it gives $h_{gw} \approx 11$ W/m²K (vs a fixed 15). It falls back to 15 on any CoolProp failure or a negligible gas/wall ΔT . Because $\approx 11 < 15$, the calculated value puts slightly *less* heat into the gas, so it does not close the gas-temperature gap (below) - it isolates that gap as stratification rather than the wall coefficient.

6.5 Gas-liquid interphase

`calculation.h_gas_liquid` accepts “calc”/“calc_two_sided” (natural-convection correlations across the interface) or a fixed number. For a liquid drain a small fixed value (≈ 2 W/m²K) keeps the warm gas decoupled from the cold boiling liquid (near-adiabatic), reproducing the measured warm gas / cold liquid split.

6.6 Initialisation of the warm gas and split wall

Two related initialisation features (both keyed on `initial_gas_temperature`):

- **Superheated gas zone.** With `initial_gas_temperature` the gas zone starts warmer than the saturated liquid (a lower density and mass; the fill takes up the difference at a matched total mass). The value is set to the **mid-band (median) of the top 3 gas-space thermocouples** at t_0 - a representative warm-gas start rather than the single warmest sensor.
- **Split wall init.** With a superheated gas zone the wall has equilibrated with the phase it touches before the blowdown, so the **gas-contact wall starts at the gas temperature** and the **wetted wall at the liquid temperature** (rather than both at `initial_temperature`). An optional `initial_wall_temperature` gives a uniform override. Starting the gas and gas-wall warm lifts the modelled gas by $\sim 0.4\text{--}5^\circ\text{C}$ toward the measured top, with the wetted wall, retained mass and rates unchanged. Default behaviour (no `gas_temperature`) is unchanged.

6.7 Zone mass and energy balances (NEM)

Each zone integrates its own internal energy. The outflow enthalpy is **attributed to the phase the mass actually leaves from**: a liquid-space release draws liquid, a vapour-space release / standard blowdown draws gas (exactly one of h_g^{out} , h_l^{out} is non-zero):

$$\begin{aligned}\Delta U_g &= -\dot{m} h_g^{\text{out}} \Delta t + \dot{Q}_{\text{inner}} \Delta t - \dot{Q}_{gl} \Delta t + E_{\text{evap}} - E_{\text{cond}} \\ \Delta U_l &= -\dot{m} h_l^{\text{out}} \Delta t + \dot{Q}_{\text{inner}}^w \Delta t + \dot{Q}_{gl} \Delta t - E_{\text{evap}} + E_{\text{cond}}\end{aligned}\quad (6.5)$$

where $\dot{Q}_{gl} = h_{gl} A_{\text{interface}} (T_g - T_l)$ is the interphase heat and E_{evap} , E_{cond} are the boil-off / condensation phase-transfer energies handled in the mass balance. Both wall $\rightarrow$ gas ($\dot{Q}_{\text{inner}}$) and wall $\rightarrow$ liquid ($\dot{Q}_{\text{inner}}^w$) terms are present, so the wall heats both zones.

6.8 The gas-temperature gap (stratification)

For the **gas** releases the modelled 0-D gas node runs colder than the measured warm top (e.g. ~ -49 vs -27°C for test 5). The real gas is **stratified** - warm on top, cold near the interface - which a single 0-D gas node cannot represent; it averages the two and mixes in cold boil-off vapour. This is a fundamental 0-D limitation, not a heat-transfer-coefficient error (the wall model reproduces the measured wall, and the gas-wall HTC is the physical natural-convection value). Capturing it would need a stratified (multi-node) gas zone. The warm-gas initialisation above reduces, but does not remove, the gap.

Fig. 6.1 shows the mechanism for a gas release: the gas expansion-cools during the fast blowdown, then **re-warms** during the constant-pressure plateau as the wall heats the near-isobaric gas, while the two wall nodes separate into a warm gas-contact wall and a cold wetted (dry-ice-contact) wall.

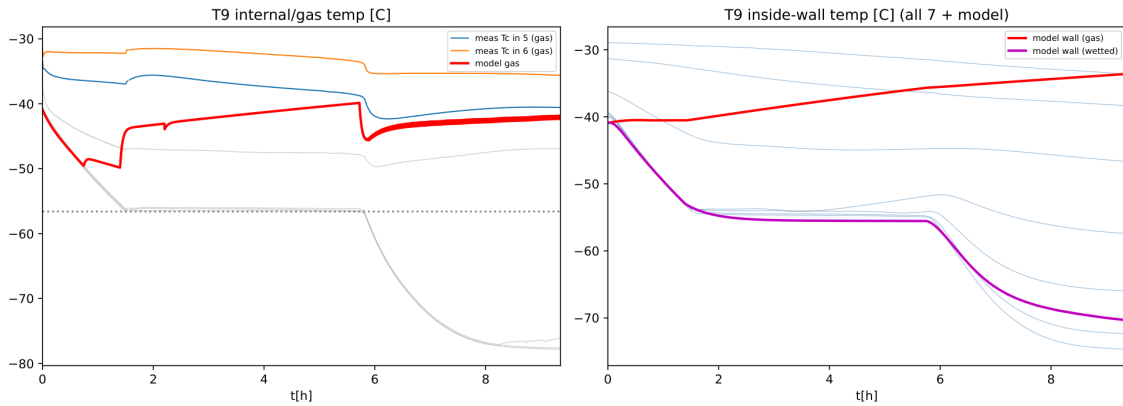


Fig. 6.1: Test 9 gas-release diagnostic: internal/gas temperatures (left) and inner-wall sensors vs the two modelled wall nodes (right).

VALIDATION

The CO₂ add-on is validated against the full pure-CO₂ CARDICE set (tests 5-10, *Experimental Basis (CARDICE)*) using the open 1 Hz Ineris data [6, 8]. Each case has a tracked input file in `validation/CARDICE_*.yaml` and a four-panel comparison figure.

7.1 Methodology

For every test:

1. **Time base** - the blowdown start t_0 is detected from the onset of the load-cell mass decline (and the pressure drop); all data are shifted to it.
2. **Initial mass** - `vessel.liquid_level` is calibrated (via the CoolProp initial phase masses) to the measured load-cell inventory. All six match within 0.4 %.
3. **Discharge** - a single physical $C_d = 0.68$; the orifice is inferred from the reliable gas discharge and the liquid gets the HNE boost N (*Discharge Model*).
4. **Gas start** - `initial.gas_temperature` = mid-band of the top 3 gas-space thermocouples; the gas-contact wall starts at that temperature and the wetted wall at the liquid temperature (*Heat and Mass Transfer*).
5. **Heat transfer** - $h_{\text{outer}} = 0.4 \text{ W/m}^2\text{K}$ (Jamois); below the triple point the gas-wall uses `solid_h_gas_wall: "calc"` (natural convection) and the wetted wall `solid_h_inner = 150 \text{ W/m}^2\text{K}`.

7.2 Master comparison

Table 7.1: Model vs measured (CARDICE tests 5-10)

Test	Phase	m_0 mdl/meas [kg]	Retained dry ice mdl/meas [kg]	Wetted (bottom) wall mdl/meas [°C]	Duration mdl/meas [h]
5	gas	825 / 823	269 / 217	−78 / −75	9.8 / 10.9
6	liquid	805 / 802	4.4 / 4.4	−25 / −24	2.7 / 2.9
7	gas	790 / 791	312 / 298	−75 / −75	6.0 / 5.9
8	liquid	760 / 758	4.6 / 3.7	−32 / −32	1.33 / 1.33
9	gas	904 / 904	402 / 412	−71 / −75	7.6 / 6.7
10	liquid	745 / 743	4.8 / ~ 0	−42 / −40	2.9 / 2.9

The central physics is reproduced across the set: **gas releases retain 270-400 kg of in-vessel dry ice**

with multi-hour triple-point plateaus and walls to -75°C ; liquid releases retain essentially none, with warm walls - the dry ice a liquid leak makes appears downstream instead. Initial mass, discharge rates (single C_d), retained mass and wall temperatures all agree.

7.3 Per-test comparisons

Each figure has four panels: vessel pressure; inventory mass (with the modelled gas / liquid / solid breakdown); all 6 internal fluid thermocouples vs the modelled gas and liquid/solid temperatures; and all 7 inner-wall thermocouples vs the modelled gas-contact and wetted wall nodes.

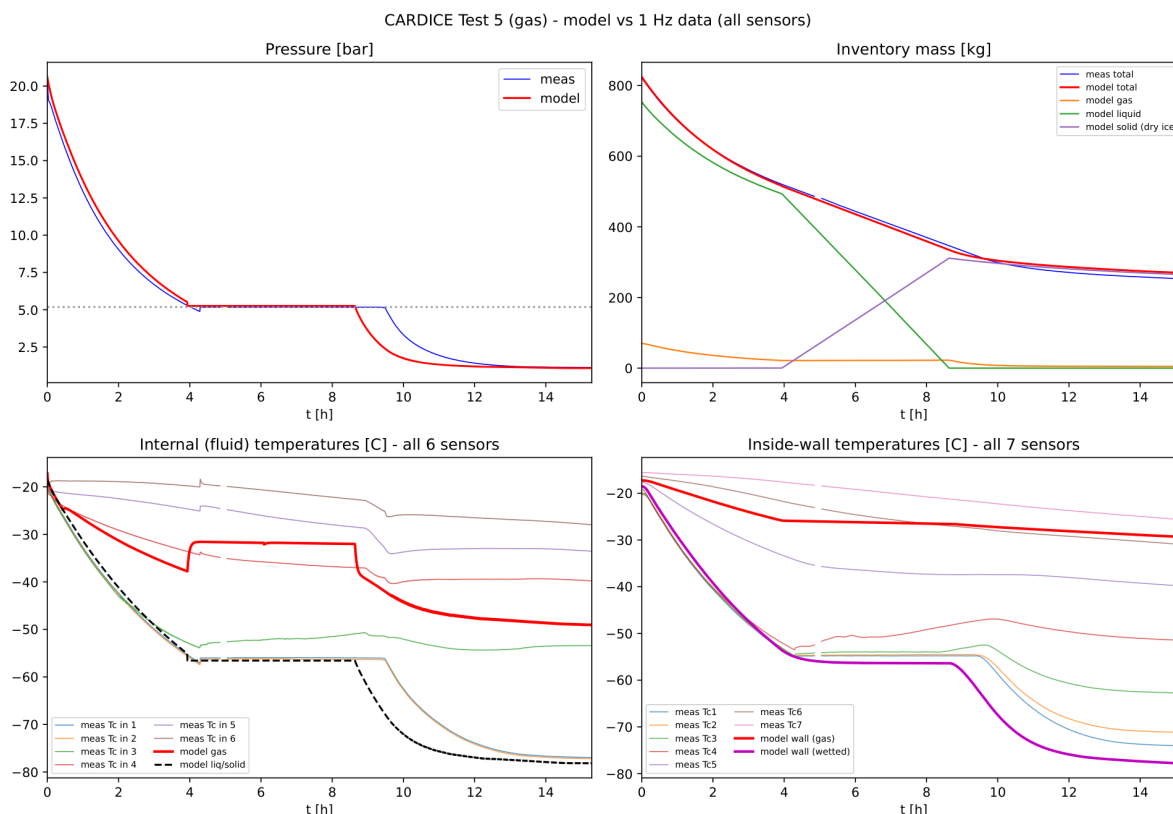


Fig. 7.1: Test 5 - gas release, 20 bar. Long triple-point plateau, ~ 270 kg retained dry ice, wetted wall to -78°C .

7.4 Benchmark against Vessfire (Test 5)

Test 5 is the paper's benchmark. [6] report the measured pressure, mass flow rate, phase temperatures and wall temperatures against Vessfire (Fig. 7.7). HydDown reproduces the same three-stage behaviour - gas/liquid blowdown, triple-point plateau, gas/solid sublimation descent - and, like Vessfire, captures the pressure and rate well while under-predicting the (stratified) gas temperature.

7.5 Sensitivity: wall-to-solid coefficient

The wetted (bottom) wall minimum below the triple point is set by `solid_h_inner`. A sweep on test 7 shows it approaches the -76°C solid asymptote; $150 \text{ W/m}^2\text{K}$ matches the measured -75°C coldest sensor.

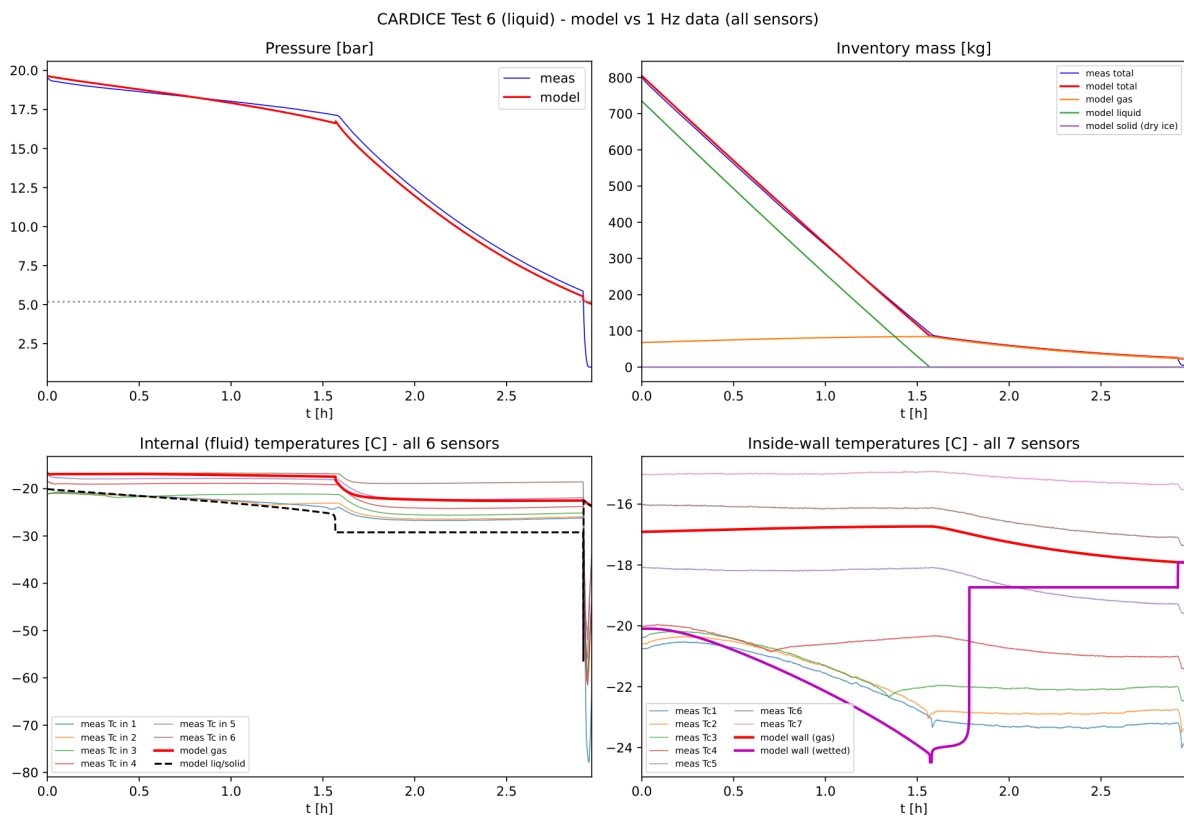


Fig. 7.2: Test 6 - liquid release, 20 bar. Liquid drains and boils out before the triple point; no in-vessel dry ice; pressure and inventory track the data to ~ 8 bar / 2.5 h.

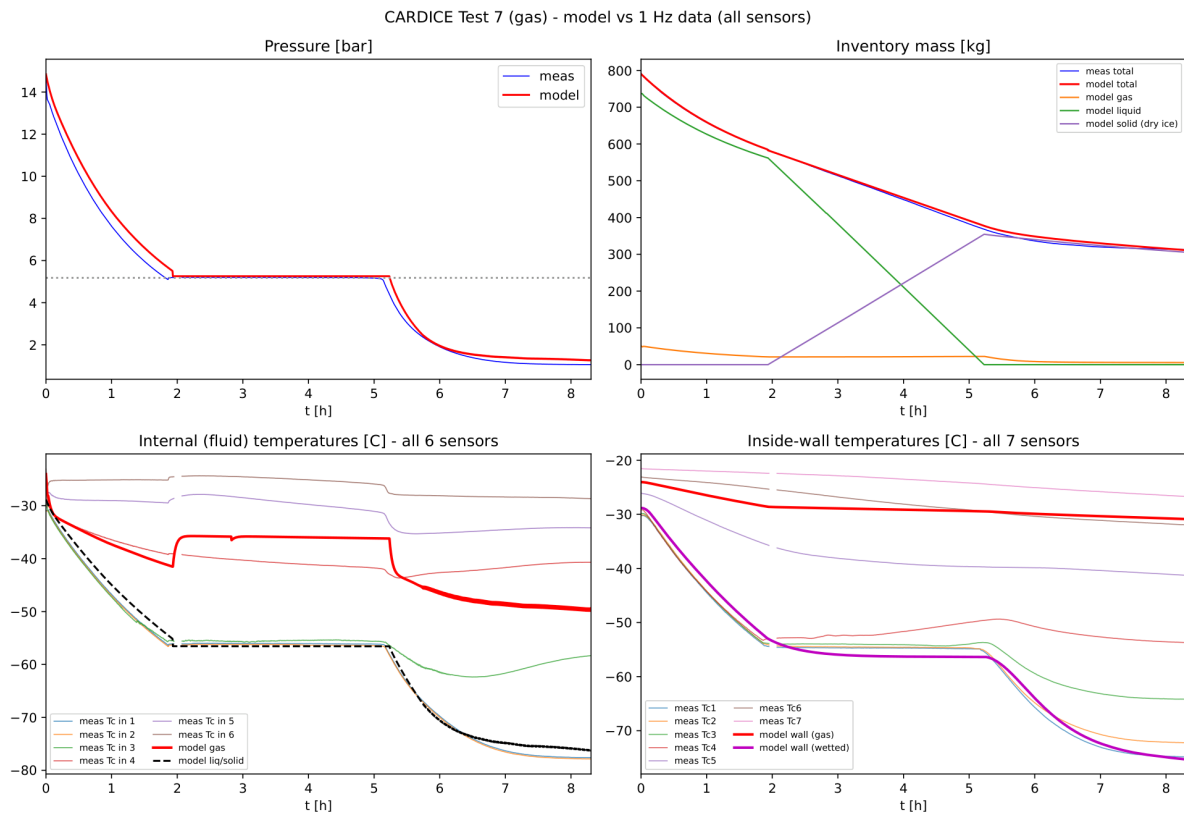


Fig. 7.3: Test 7 - gas release, 15 bar.

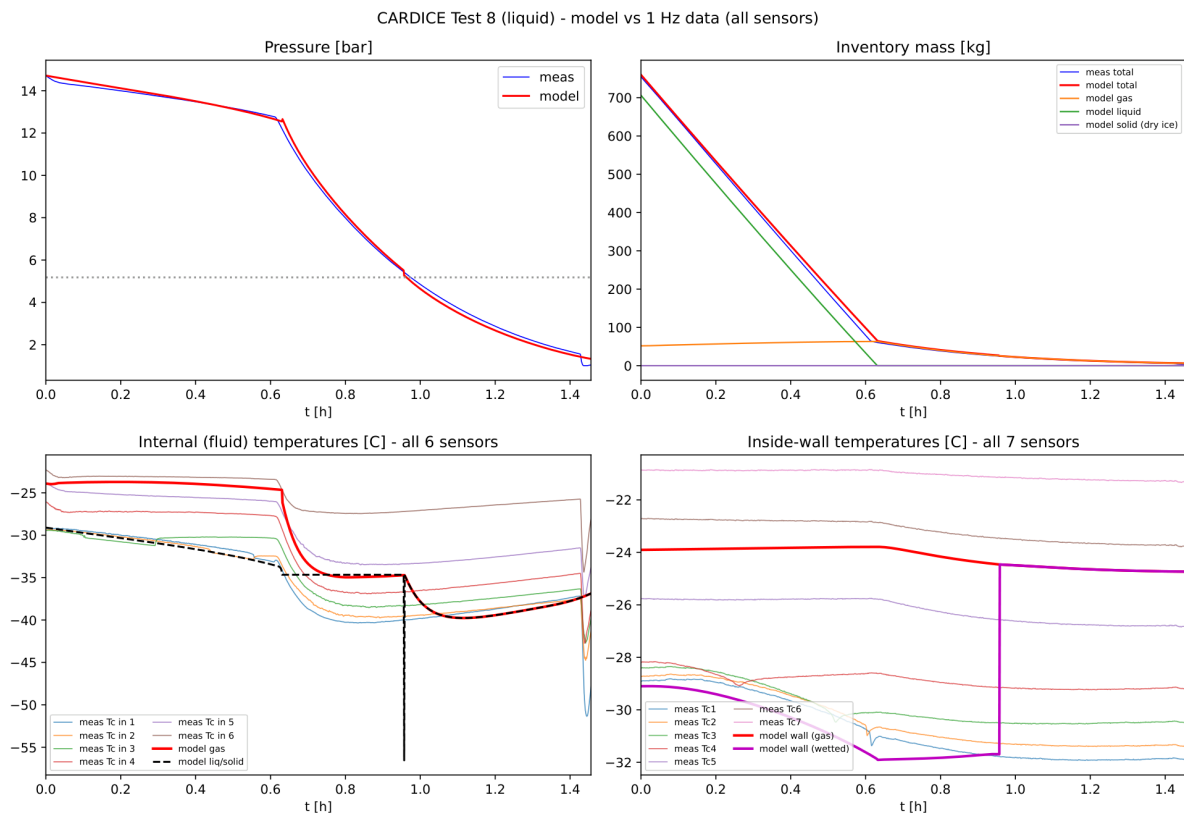


Fig. 7.4: Test 8 - liquid release, 15 bar. Single C_d : liquid 0.311 vs 0.317, gas tail 0.030 vs 0.030 kg/s, duration 1.33 vs 1.33 h.

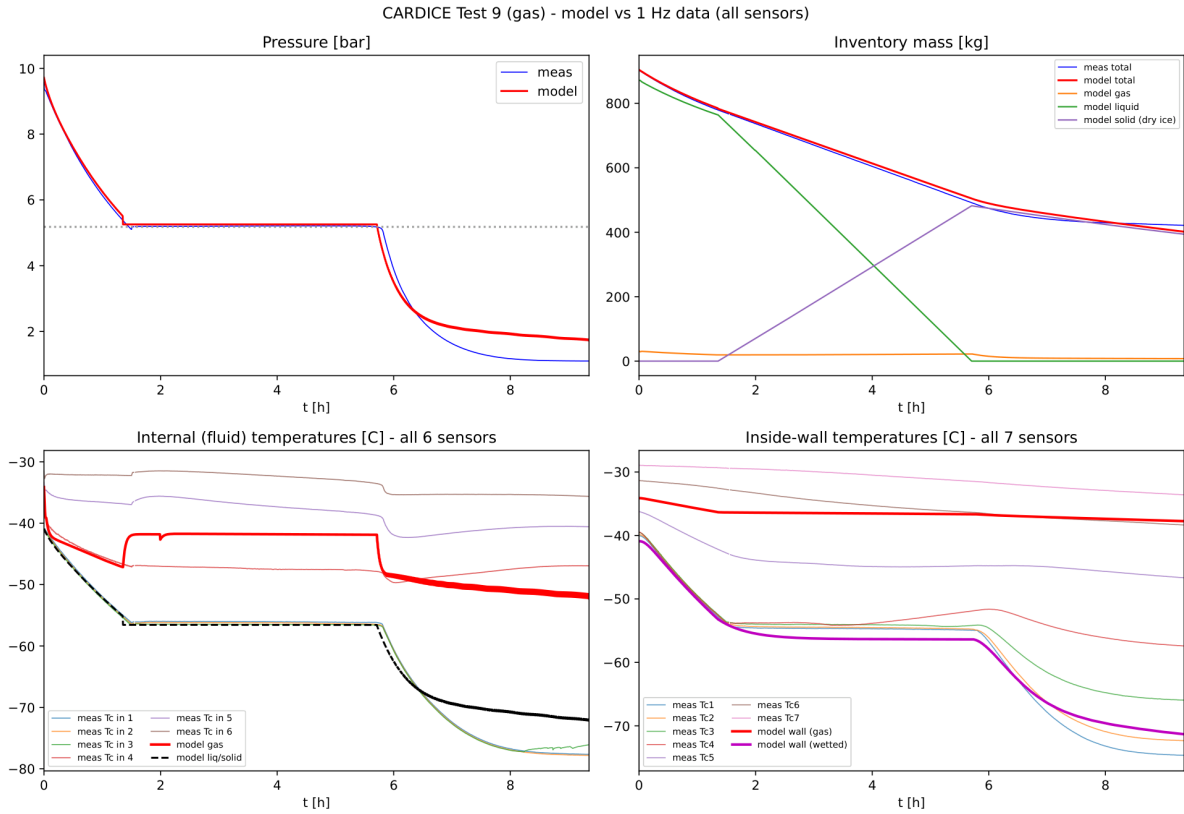


Fig. 7.5: Test 9 - gas release, 10 bar. The most retained dry ice (~ 400 kg).

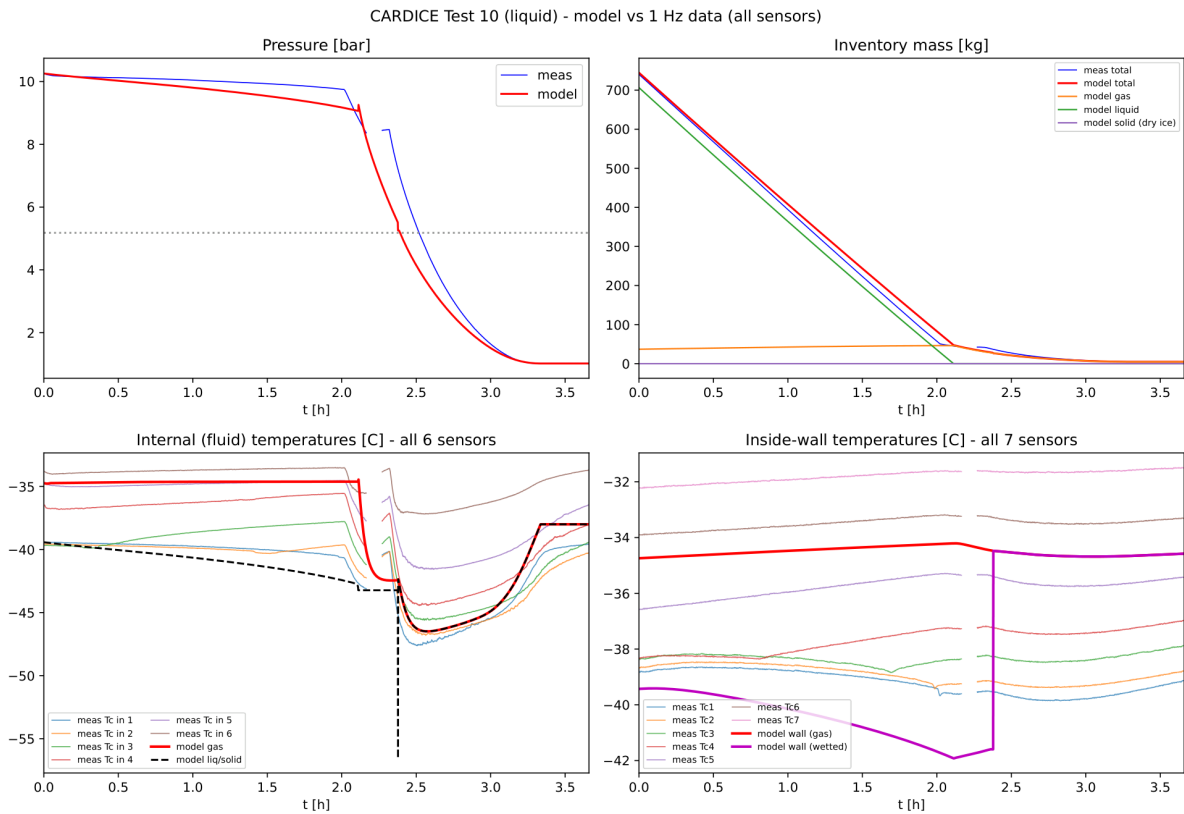


Fig. 7.6: Test 10 - liquid release, 10 bar. Orifice from the liquid rate (4.09 mm); duration 2.89 vs 2.92 h.

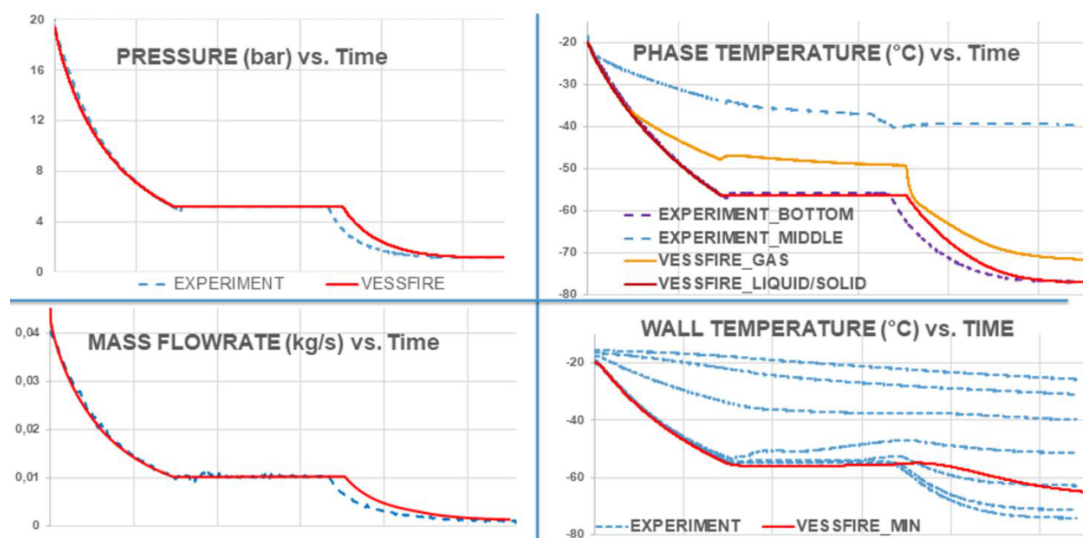


Fig. 5. Test 5 – Experimental vs. simulation results.

Fig. 7.7: Test 5 experimental vs Vessfire results (pressure, mass flow rate, phase temperature, wall temperature).

From [6].

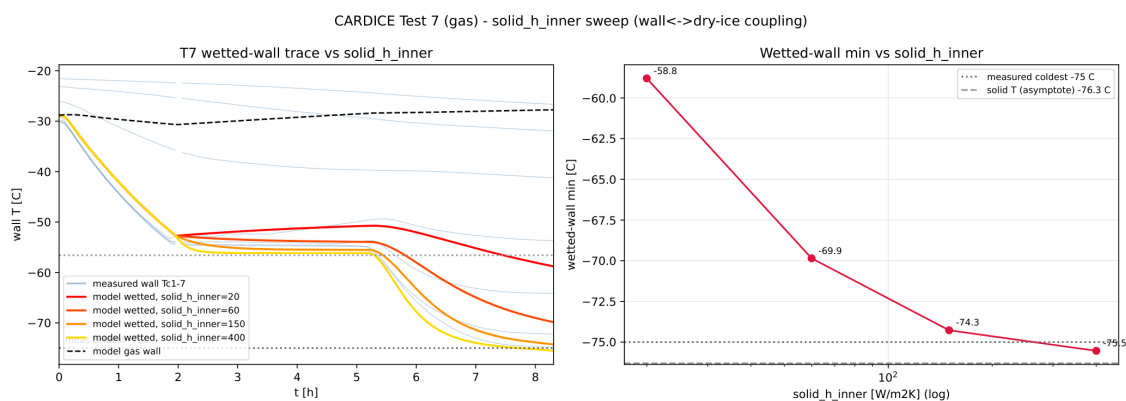


Fig. 7.8: Test 7 wetted-wall temperature and its minimum vs solid_h_inner.

7.6 Remaining gaps

These are documented model limitations, not defects:

- **Gas-temperature stratification.** The 0-D gas node runs colder than the measured warm top on the gas cases (~ -49 vs -27 °C); a single node cannot hold the vertical gradient (*Heat and Mass Transfer*). Vessfire, a 0-D-gas tool, shows the same [6].
- **Triple-point handover kink.** A ~ 3 °C gas-temperature kink at the CoolProp $\rightarrow$ thermopack vessel hand-over (liquid cases); small vs the stratification gap; deferred.
- **Test 6 abrupt final collapse.** The measured vessel empties abruptly at ~ 2.9 h - faster than any physical orifice for the residual gas - a non-orifice end-effect (blowdown-valve full-open / final venting) that a steady discharge model does not reproduce.
- **Gas vs liquid orifice at 10 bar (test 10).** The gas- and liquid-inferred orifices differ by ~ 8 % because the equilibrium HEM liquid over-predicts at 10 bar; see the phase-specific- C_d note in *Discharge Model*.

7.7 Regression

The base HydDown behaviour is unchanged: all new inputs default off (`liquid_nonequilibrium: 0`, no `gas_temperature`, fixed `solid_h_gas_wall`), and `test_all.py` passes except two pre-existing, unrelated failures (a Tk/plotting environment issue and a modified `rupture.yml` example). A dedicated `src/hyddown/test_co2_release.py` covers the CO₂ module.

**CHAPTER
EIGHT**

BIBLIOGRAPHY

BIBLIOGRAPHY

- [1] Anders Andreasen. HydDown: a python package for calculation of hydrogen (or other gas) pressure vessel filling and discharge. *Journal of Open Source Software*, 6(66):3695, 2021. doi:10.21105/joss.03695.
- [2] Ian H. Bell, Jorrit Wronski, Sylvain Quoilin, and Vincent Lemort. Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp. *Industrial & Engineering Chemistry Research*, 53(6):2498–2508, 2014. doi:10.1021/ie4033999.
- [3] Roland Span and Wolfgang Wagner. A new equation of state for carbon dioxide covering the fluid region from the triple-point temperature to 1100 K at pressures up to 800 MPa. *Journal of Physical and Chemical Reference Data*, 25(6):1509–1596, 1996. doi:10.1063/1.555991.
- [4] Øivind Wilhelmsen, Ailo Aasen, Geir Skaugen, Peder Aursand, Anders Austegard, Eskil Aursand, Magnus A. Gjennestad, Halvor Lund, Gaute Linga, and Morten Hammer. Thermodynamic modeling with equations of state: present challenges with established methods. *Industrial & Engineering Chemistry Research*, 56(13):3503–3515, 2017. doi:10.1021/acs.iecr.7b00317.
- [5] Yohann Le Guennec, Romain Privat, and Jean-Noël Jaubert. Development of the translated-consistent tc-PR and tc-RK cubic equations of state. *Fluid Phase Equilibria*, 429:301–312, 2016. doi:10.1016/j.fluid.2016.09.003.
- [6] G. Vaillant, L. Teberikler, A. Fahmi, F. Garcia, C. Merat, and M. Drescher. Vessel blowdown of CO₂-rich fluids: experiments and simulations. In *Proceedings of the 15th International Conference on Greenhouse Gas Control Technologies (GHGT-15)*. Abu Dhabi, UAE, 2021. SSRN preprint 3823308. URL: <https://ssrn.com/abstract=3823308>.
- [7] Didier Jamois, Christophe Proust, Leyla Teberikler, and Adil Fahmi. An experimental setup for investigating the blowdown of liquid CO₂ down to -50°C . *International Journal of Greenhouse Gas Control*, 129:103974, 2023. doi:10.1016/j.ijggc.2023.103974.
- [8] Didier Jamois and Christophe Proust. CARDICE CO₂ blowdown experiments — 1 Hz dataset. Zenodo, 2024. doi:10.5281/zenodo.17257910.
- [9] J. E. Dyer, D. B. Doran, and C. Musso. Non-equilibrium and non-homogeneous effects on the flow of flashing liquids. *Institution of Chemical Engineers Symposium Series*, 2007.
- [10] H. K. Fauske and M. Epstein. Source term considerations in connection with chemical accidents and vapour cloud modelling. *Journal of Loss Prevention in the Process Industries*, 1:75–83, 1988. doi:10.1016/0950-4230(88)80020-9.
- [11] R. C. Lopes, A. Z. Francesconi, and S. S. V. Vianna. Modelling of source term from accidental release of pressurised CO₂. *Process Safety and Environmental Protection*, 113:88–96, 2018. doi:10.1016/j.psep.2017.08.046.

- [12] J. C. Leung. A generalized correlation for one-component homogeneous equilibrium flashing choked flow. *AIChE Journal*, 32(10):1743–1746, 1986. doi:[10.1002/aic.690321018](https://doi.org/10.1002/aic.690321018).
- [13] Christie J. Geankoplis. *Transport Processes and Unit Operations*. Prentice-Hall, 3rd edition, 1993.